

Model Assessment: Predicting density and assessing factors driving marbled murrelet distribution in Puget Sound using hierarchical distance sampling

Code by S.J. Gillman

Required Packages

```
library(tidyverse)
library(MCMCvis)
library(nimble)
library(ggplot2)
```

Load Data

Required Files Previously Made or Provided

- Survey_Grid_Info_NEW.RData
- chain_1.rds
- chain_2.rds
- chain_3.rds
- nimConstants.rds
- nimData.rds
- gof_SummaryStats.rds
- N_posterior_gof.rds
- rN_posterior_gof.rds
- pcap_posterior_gof1-11.rds
- lambda_posterior_gof1-11.rds

pcap and lambda .rds files that end in the number are the iterations, 1 = 1 to 1000, 2 = 1001 to 2000, 3 = 2001 to 3000, etc. b,c,d are the PSU-Segments 2001 to 8095 broken up into 2k groups due to overall large size and needing to load to github.

```
load("../DataAnalysis/DataFiles/Survey_Grid_Info_NEW.RData")
samples_1 <- readRDS("RESULTS/COMPRESSED/chain_1.rds")
samples_2 <- readRDS("RESULTS/COMPRESSED/chain_2.rds")
samples_3 <- readRDS("RESULTS/COMPRESSED/chain_3.rds")

# Model input data
nimConstants <- readRDS("RESULTS/COMPRESSED/nimConstants.rds")
nimData <- readRDS("RESULTS/COMPRESSED/nimData.rds")

# Summarized GoF INFO
results_gof <- readRDS("RESULTS/GOF/COMPRESSED/gof_SummaryStats.rds")
```

```

N_sub <- readRDS("RESULTS/COMPRESSED/GOF/N_posterior_gof.rds")
rN_sub <- readRDS("RESULTS/COMPRESSED/GOF/rN_posterior_gof.rds")

lambda_sub <- rbind(
  cbind(
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof1.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof1b.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof1c.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof1d.rds")),
  cbind(
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof2.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof2b.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof2c.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof2d.rds")),
  cbind(
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof3.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof3b.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof3c.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof3d.rds")),
  cbind(
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof4.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof4b.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof4c.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof4d.rds")),
  cbind(
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof5.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof5b.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof5c.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof5d.rds")),
  cbind(
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof6.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof6b.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof6c.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof6d.rds")),
  cbind(
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof7.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof7b.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof7c.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof7d.rds")),
  cbind(
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof8.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof8b.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof8c.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof8d.rds")),
  cbind(
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof9.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof9b.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof9c.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof9d.rds")),
  cbind(
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof10.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof10b.rds"),

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    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof10c.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof10d.rds")),
cbind(
  readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof11.rds"),
  readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof11b.rds"),
  readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof11c.rds"),
  readRDS("RESULTS/COMPRESSED/GOF/lambda_posterior_gof11d.rds")))

pcap_sub <- rbind(
  cbind(
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof1.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof1b.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof1c.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof1d.rds")),
  cbind(
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof2.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof2b.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof2c.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof2d.rds")),
  cbind(
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof3.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof3b.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof3c.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof3d.rds")),
  cbind(
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof4.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof4b.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof4c.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof4d.rds")),
  cbind(
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof5.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof5b.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof5c.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof5d.rds")),
  cbind(
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof6.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof6b.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof6c.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof6d.rds")),
  cbind(
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof7.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof7b.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof7c.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof7d.rds")),
  cbind(
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof8.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof8b.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof8c.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof8d.rds")),
  cbind(
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof9.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof9b.rds"),

```

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    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof9c.rds"),
    readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof9d.rds")),
cbind(
  readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof10.rds"),
  readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof10b.rds"),
  readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof10c.rds"),
  readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof10d.rds")),
cbind(
  readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof11.rds"),
  readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof11b.rds"),
  readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof11c.rds"),
  readRDS("RESULTS/COMPRESSED/GOF/pcap_posterior_gof11d.rds")))

chains <- coda::mcmc.list(samples_1,samples_2,samples_3)

result_summary <- MCMCsummary(chains, probs = c(0.025,0.25,0.5,0.75, 0.975)) %>%
  rownames_to_column("parameter") %>%
  mutate(
    variable = sub("\\[.*", "", parameter),
    model_id = gsub("\\\\", "", sub(".*\\\\[", "", parameter)))

round(max(result_summary$Rhat, na.rm =T),2)

## [1] 1

pd_dir <- bayestestR::p_direction(chains)

round(max(results_gof$Rhat, na.rm =T),2)

## [1] 1.08

```

N GoF

```

# result matrix
N_new <- matrix(0, nrow = nimConstants$nS, ncol = nrow(N_sub))

# Loop over sites
for (r in 1:nimConstants$nS) {
  my_idx <- nimConstants$MONTHYEAR[r]

  # Extract vectors for all iterations at once
  rN_vec <- rN_sub[, my_idx]
  lambda_vec <- lambda_sub[, r]

  # Calculate p
  prob_p <- rN_vec / (rN_vec + lambda_vec)

  # Create new N estimates
  N_new[r, ] <- rbinom(nrow(N_sub), size = rN_vec, prob = prob_p)
}

```

```

# Residuals for 'observed'
FT_obs <- colSums((sqrt(t(N_sub)) - sqrt(t(lambda_sub)))^2)
# Residuals for 'new'
FT_sim <- colSums((sqrt(N_new) - sqrt(t(lambda_sub)))^2)

# Posterior Predictive P-value
round(mean(FT_sim > FT_obs),2)

## [1] 0.88

```

Counts GoF

```

C_actual <- nimData$C

C_new <- matrix(0, nrow = nimConstants$nS, ncol = nrow(N_sub))
C_expected <- matrix(0, nrow = nimConstants$nS, ncol = nrow(N_sub))

for (r in 1:nimConstants$nS) {
  N_vec <- N_sub[, r]
  pcap_vec <- pcap_sub[, r]

  # expected Counts
  C_expected[r,] <- N_sub[, r] * pcap_sub[, r]
  # Create new C estimates
  C_new[r, ] <- rbinom(nrow(N_sub), size = N_vec, prob = pcap_vec)
}

# Residuals for 'observed'
FT_obs <- colSums((sqrt(nimData$C) - sqrt(C_expected))^2)
# Residuals for 'new'
FT_sim <- colSums((sqrt(C_new) - sqrt(C_expected))^2)

#Posterior Predictive P-value
round(mean(FT_sim > FT_obs), 2)

## [1] 0.54

```

Prior posterior overlap

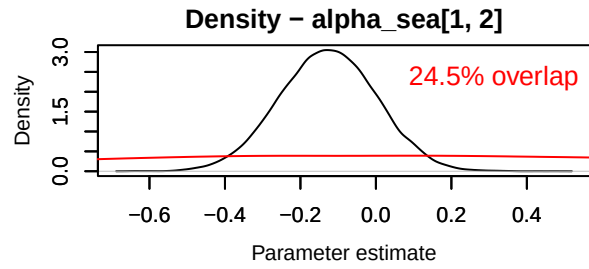
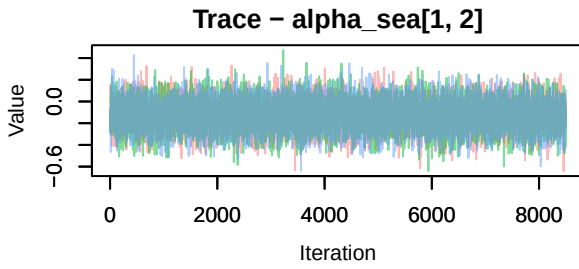
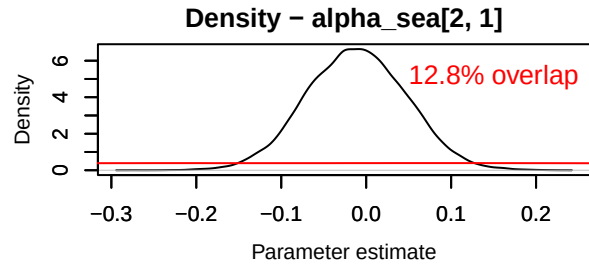
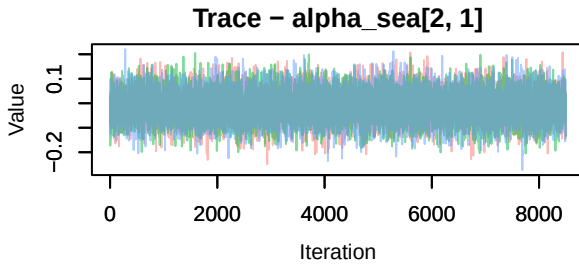
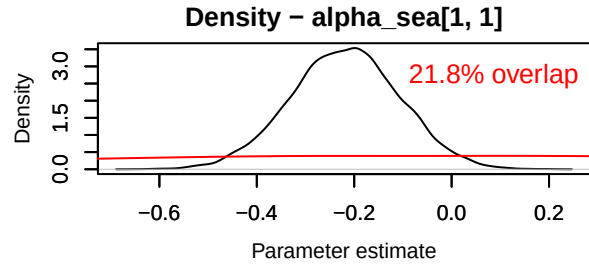
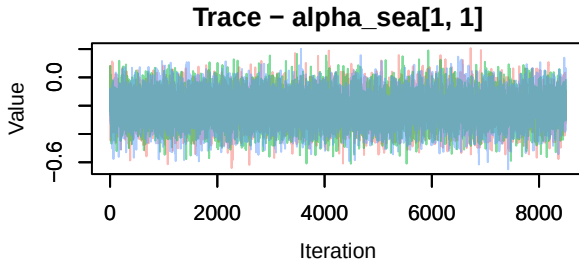
```

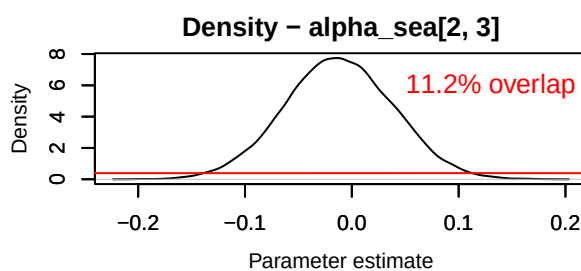
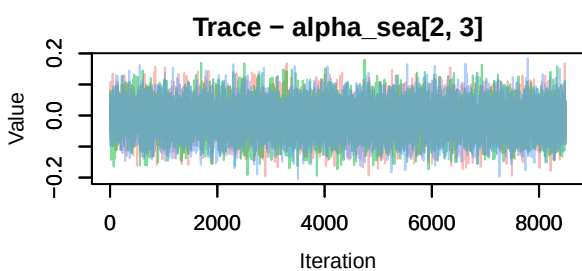
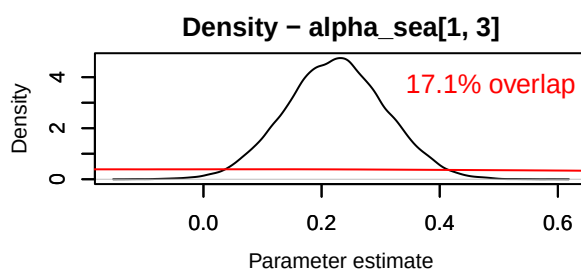
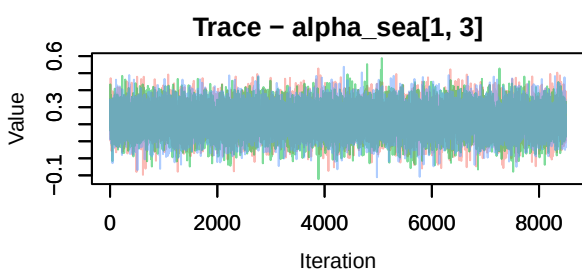
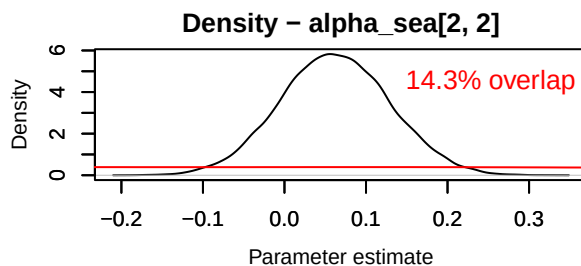
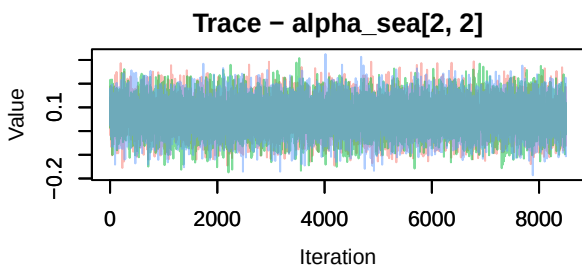
PR <- rnorm(25500, 0, 1)

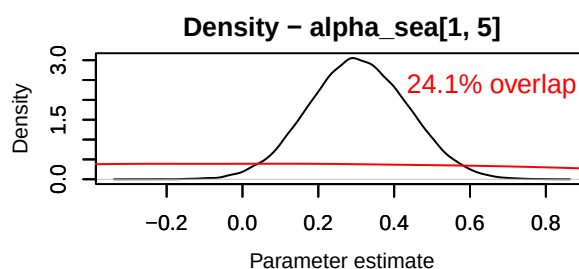
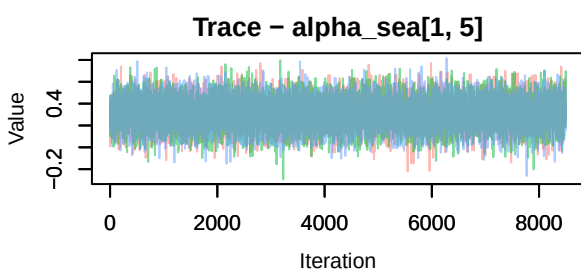
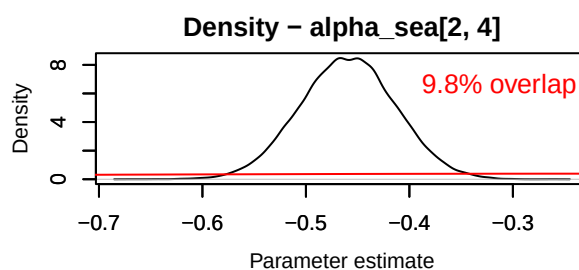
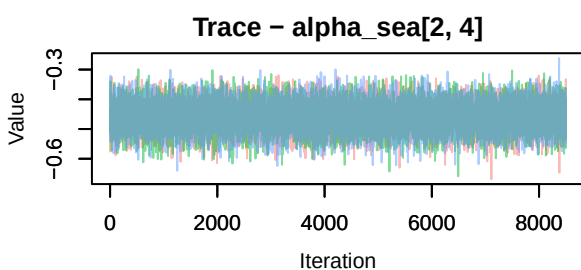
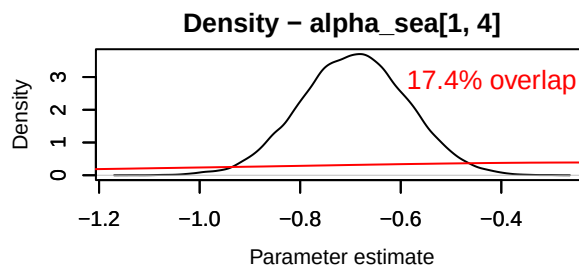
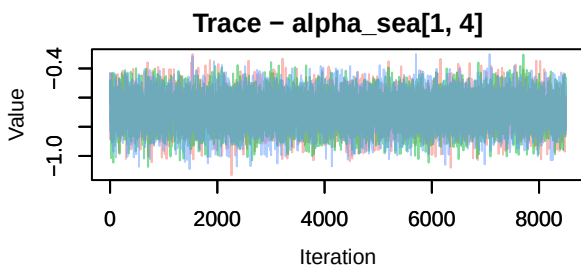
MCMCtrace(chains, params = 'alpha_sea', iter = 8500, priors = PR, pdf = FALSE)

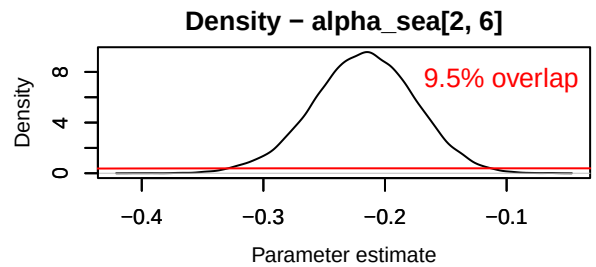
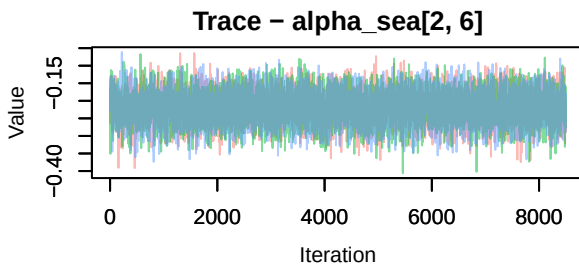
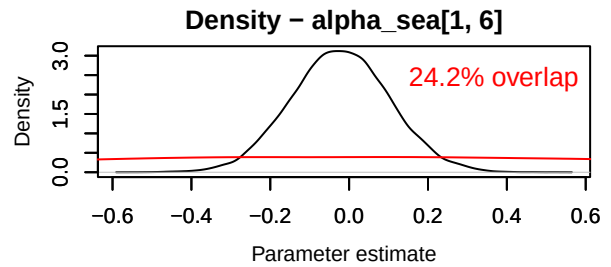
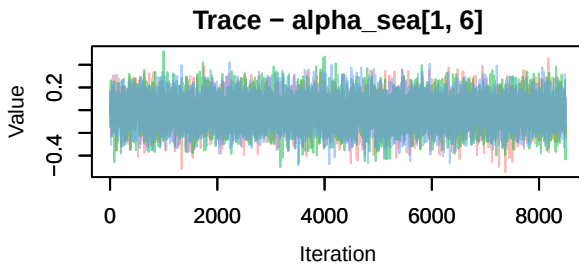
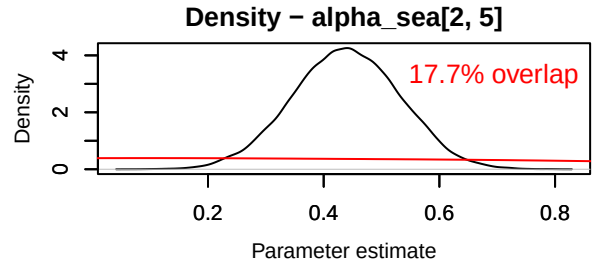
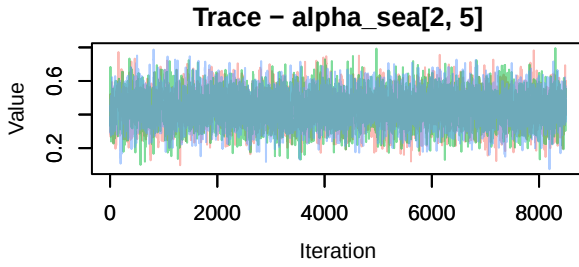
## Warning in MCMCtrace(chains, params = "alpha_sea", iter = 8500, priors = PR, :
## Only one prior specified for > 1 parameter. Using a single prior for all
## parameters.

```









```
MCMCtrace(chains, params = 'alpha_stat', iter = 8500, priors = PR, pdf = FALSE)
```

```
## Warning in MCMCtrace(chains, params = "alpha_stat", iter = 8500, priors = PR, :
## Only one prior specified for > 1 parameter. Using a single prior for all
## parameters.
```

